



Behind Enemy Lines:

Engaging and Disrupting
Ransomware Web Panels

DEF CON 32



V I E W E R

DISCRETION

A D V I S E D

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Research interests are API for IoT and
lately C2 for malware and ransomware

 @evstykas  atropos.ai  stykas.com



HOW IT **STARTED**

Last year's DEF CON talk was on **malware** C2

Wanted a **bigger** challenge

Could it be any **harder**?

HOW IT **STARTED**



Marcus Hutchins • 1st

Threat Intelligence, Public Speaker,
Content Creator
11h • 🌐



Cyber Threat intelligence is such a wild industry. In regular intelligence the government has near total monopoly and everything is classified at TS/SCI. Whereas in CTI it'll just be some dude named Brad who got really baked one night and yolo'd his way into a major APT's backend server.



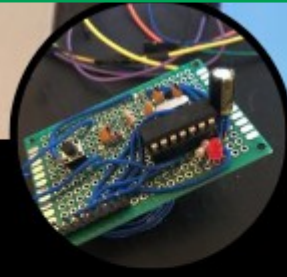
455

29 comments • 29 reposts



Solfer

@s0lfer Follows you



Iskuri

@Iskuri1 Follows you



Charles M. Ishihara

@no_th_a_n_k_s Follows you

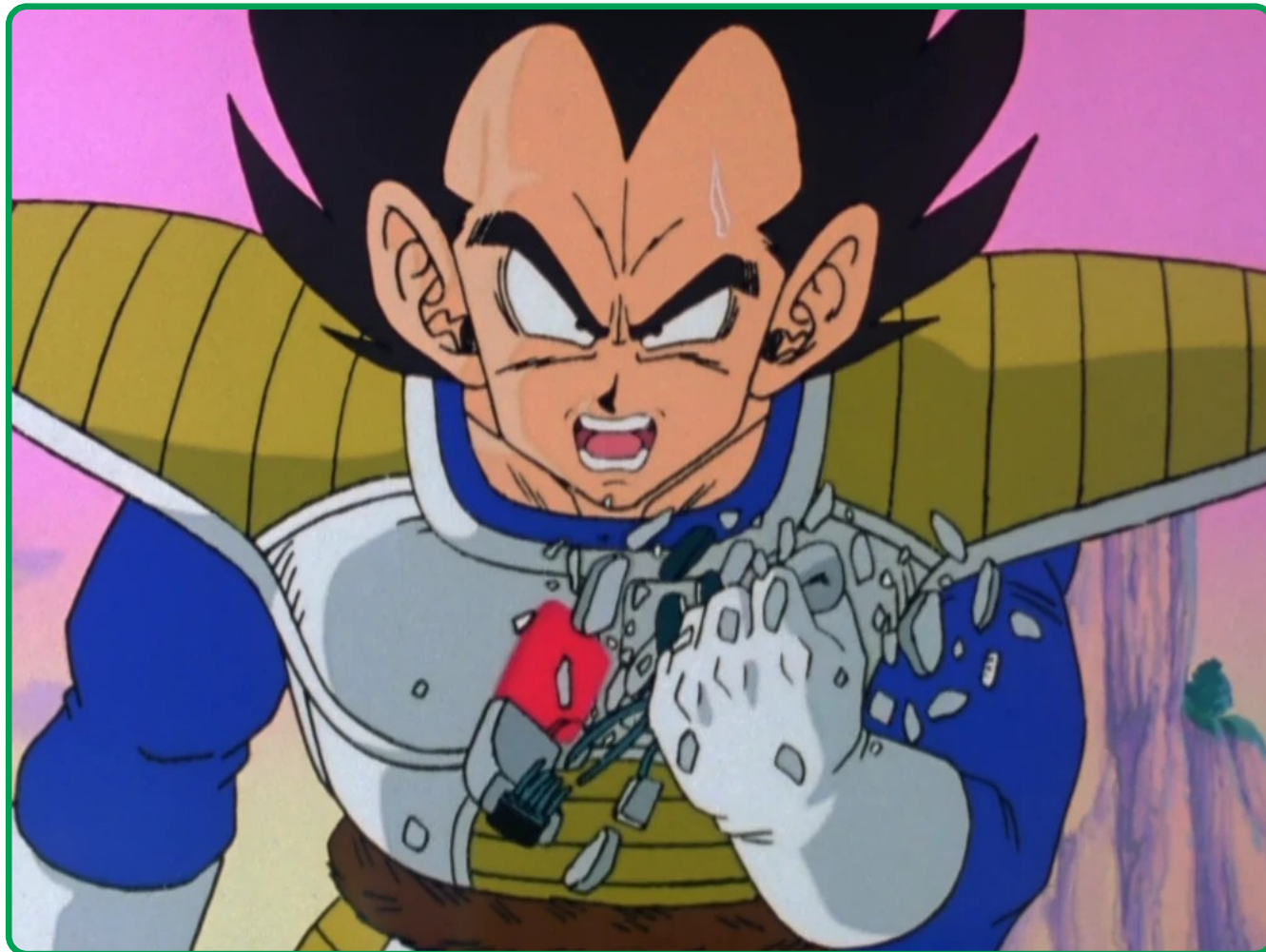
Easy? Think Again

Malware

18 out of 36

Ransomware

3 out of 140



Quick Intro to Ransomware

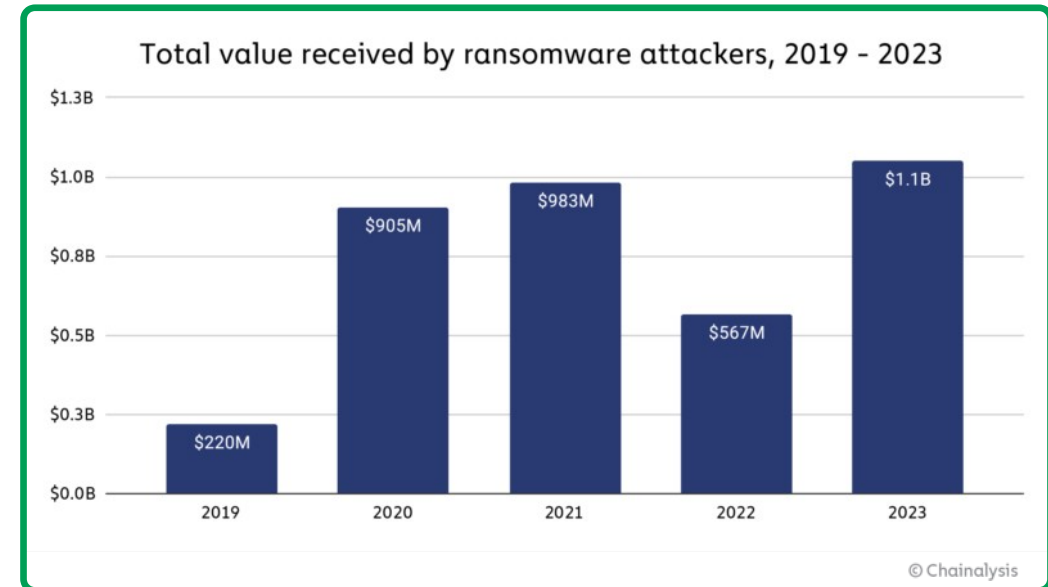
Ransomware is a type of malicious software (**malware**) that threatens to publish or blocks access to data or a computer system, usually by **encrypting** it, until the victim **pays a ransom fee** to the attacker. In many cases, the ransom demand comes with a **deadline**. If the victim doesn't pay in time, the data is **gone forever** / the ransom **increases** / moving to another way of **extortion**.

Quick Intro to Ransomware

- **Malware** Distribution and Infection
- Command and Control (**C2s**)
- Discovery and **Lateral Movement**
- Data **Extraction**
- Data **Encryption**
- **Extortion**
- Resolution

Ransomware market

- **Faster growing** type of cybercrime
- Payouts hit **\$1.1B** in 2023
- Highly **professional** industry
- Seems that boundaries are falling off...
- WannaCry opened the can of worms



Ransomware Gangs

HIGHLY Hierarchical.

Clear Structure

Tech Part

Ransom **Negotiators** and Customer Support

Money **Launderers** and Payment Processing

Collaboration and **Partnership**

Ransomware Gangs

Malware Developers

Exploitation (Possible 0 day/N-day) Developers

Data **Theft** and **Leaks**

Operational Security

Infrastructure and Hosting

Ransomware Models

Lone Wolves

Initial Access Brokers

All-in-One Ransomware Groups

RaaS

RaaS

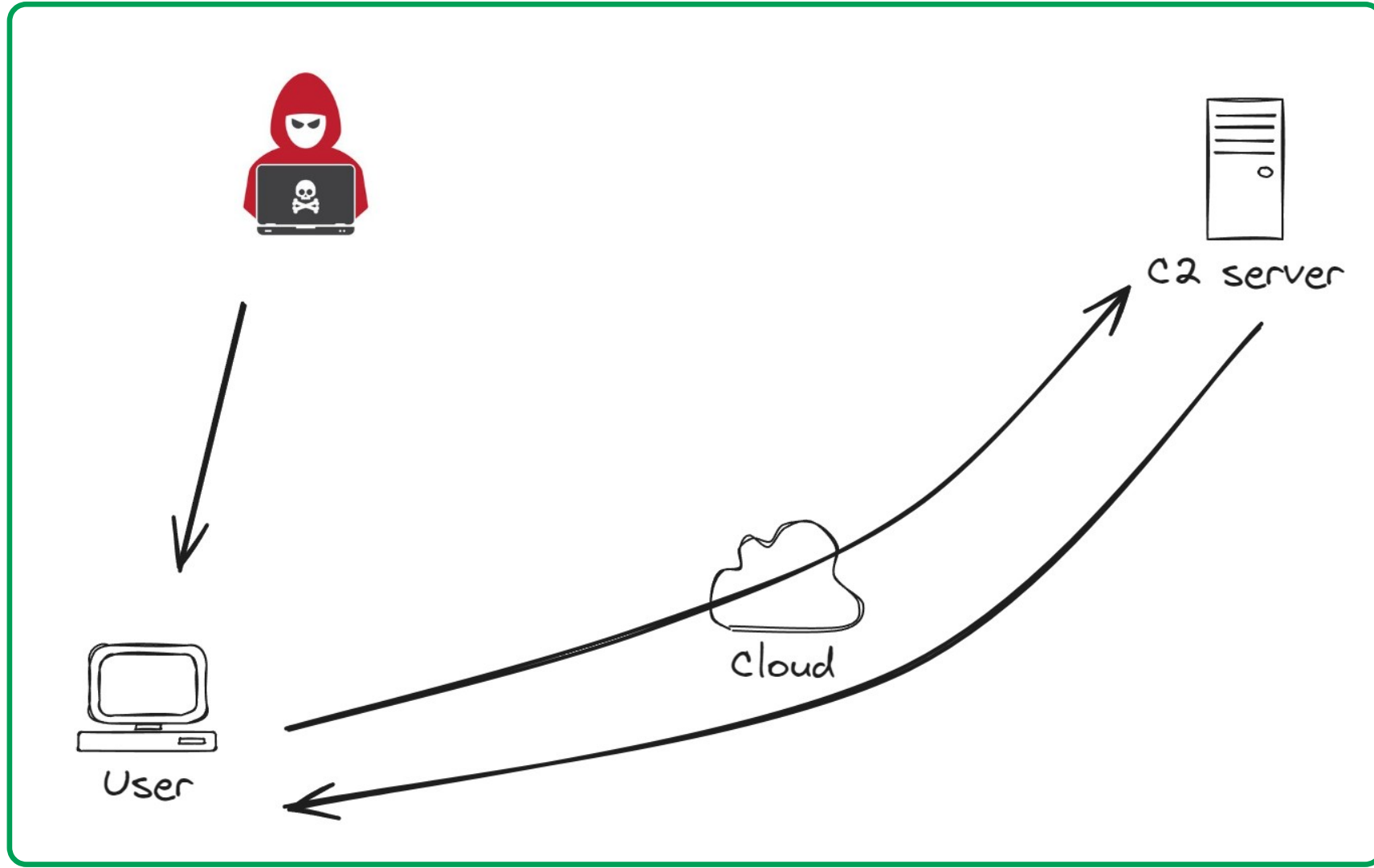
Flat **monthly fee**

Affiliate programs with a **monthly percent** of the profits

One-time license

Pure **profit sharing** / payments made to the RaaS

Most well known were **Lockbit, BlackCat, Clon** etc...



Extortion

Establish communication with victims

Lay their terms

Extort victims in multiple ways

Extortion

Ransom on data

Release data

DDoS with knowledge obtained from data

Communicate with customers / stakeholders

Let's Set Some Goals

- **Identify** C2s and data leak sites
- Try to **find vulnerabilities**
- **Identify people** behind them
- Try to **disrupt panels** / threat actors
- **DO NOT** disturb active **LEA investigations**
- **Don't be a malakas!**
- **Don't get vanned**





Government-backed attackers may be trying to steal your password

There's a chance this is a false alarm, but we believe we detected government-backed attackers trying to steal your password. This happens to less than 0.1% of all Gmail users. We can't reveal what tipped us off because the attackers will take note and change their tactics, but if they are successful at some point they could access your data or take other actions using your account. To further improve your security, based on your current settings we recommend:

Join the Advanced Protection Program

Google's strongest protection for users at risk of targeted attacks.



Identifying Panels

- <https://ransomlook.io/>
- <https://ransomware.live/>
- <https://github.com/fastfire/deepdarkCTI>
- Lots of **doom scrolling** and **monitoring** CTI companies for new posts

Limitations

C2s had a **terribly small** lifetime

Data leaks sites are **looked after**

Data leaks are behind **tor / onion**

Methodology

Ignore malware distribution and reversing

Run the malware in a sandbox

Extract all URLs

Use data-leak URLs found via CTI



Methodology

- Dirsearch (FTW) and ffuf
- Burp Suite
- Tor expert bundle , Tor Browser and torsocks
- Coffee
- ANY.RUN
- Several droplets on DigitalOcean
- Shodan.io and Censys
- Lots of coffee

Methodology

- **Black box** web app testing
- Use any **acquired information** for furthering my **attacks**
- **Interact** with the **data leak** and **chat** websites
- Intentionally **infect sandbox** or one machine to **get a ticket**

Methodology

Dirsearch and **ffuf** with a custom wordlist through **torsocks**

Only **15 URLs** gave something interesting back

Moved to **manually** check all of them

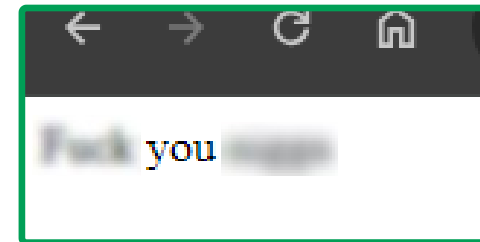
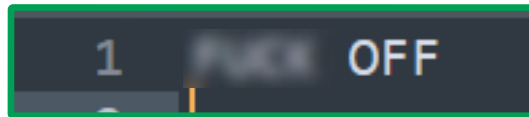
Results

5 of them were **WordPress**

A couple of them were **leaking** IP addresses

Some of them were **cheeky...**

Results



```
<footer class="text-muted py-5 bg-dark">
  <div class="container">
    <p class="float-end mb-1">Copyright © <a href="/">Abyss</a> - 2023</p>
    <p class="mb-1">Just business, nothing personal</p>
  </div>
</footer>
```




MALLOX

aka TargetCompany, FARGO and Tohnichi

First appearance **June 2021**

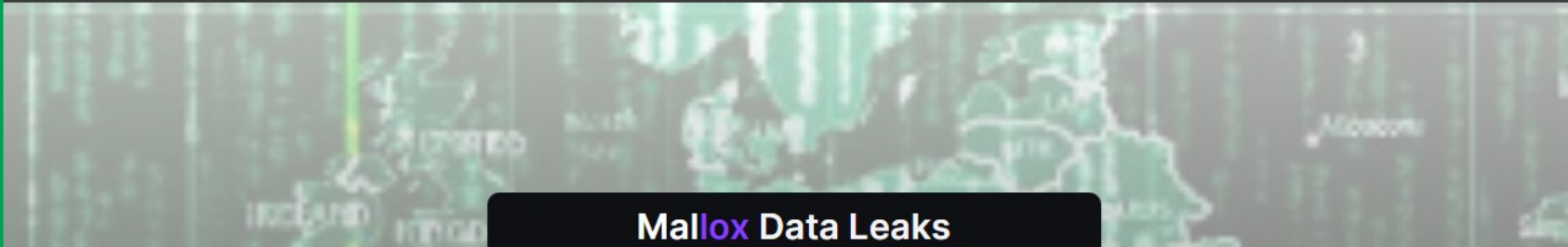
Targets mostly **Windows** machines

Exploits **MS SQL** (brute force **sa** accounts)

Hundreds of victims

MALLOX

← → ↻ ⚠ Not secure wtyafjyhwrqgo4a45wdvwwhen3cx4eue73qvlhkhvltrexjoyuklaad.onion ☆ 🌐 🍪 ⚙️ 📁 👤



Mallox Data Leaks



14/07/2024 08:32

XENAPP-GLOBER

PUBLISHED

 View



13/07/2024 18:44

integraservices

PUBLISHED

 View



05/06/2024 13:48

"Moshe Kahn Advocates"

Moshe Kahn Advocates

Moshe Kahn Advocates is a specialist business law firm based in Tel Aviv, Israel.

PUBLISHED

 View



09/04/2024 04:27

speditionlangen.de

https://www.zoominfo.com/c/heuter/426931393

https://gofile.io/d/EqTLB7

PUBLISHED

 View



17/03/2024 00:03

Rafum Group

https://pk.linkedin.com/company/rafum-group

PUBLISHED

 View



14/03/2024 10:39

Ramdev Chemical Industries

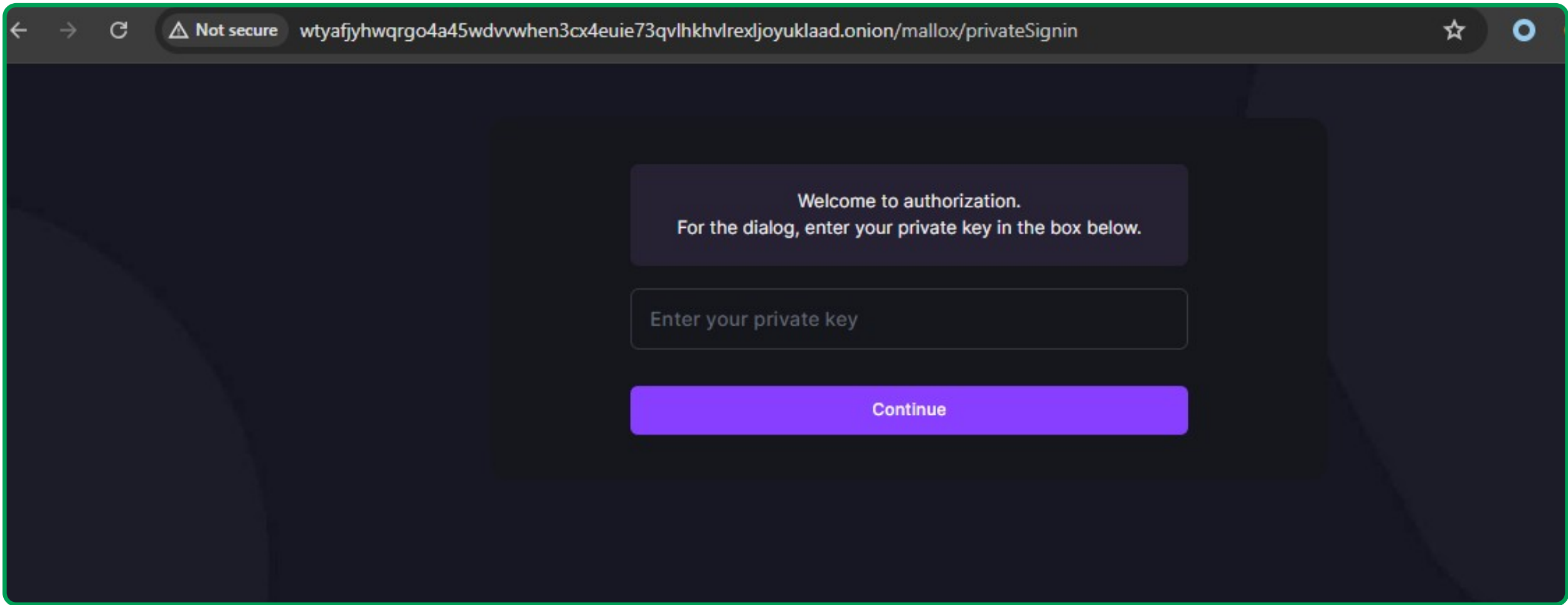
INTRODUCTION

Ramdev is promoted by a first generation technocrat Mr. Sureshkumar Bhingradia in the late 1980s and specialized in Dhthalocyanine.

PUBLISHED

 View

ATROPOS



Client information

TargetID OCB8EBFA70F48DB2BD477BD1

Your data Unpublished

Answers to FAQs

How can I trust you? ▾

Can I decrypt all files after payment? ▾

When are you usually online? I can't get a response from you? ▾

Which network should I make the payment on? ▾

How will you know that it was I who paid? ▾

How to buy bitcoins or USDT? They are banned in my country ▾

How to test my files? ▾

How to back my files? ▾

PRICE: \$20000 / 0.2959562 BTC

AMOUNT PAID: \$0

LAST TRANSACTION: N/A

[BTC] BITCOIN ADDRESS FOR PAYMENT



33i665c8Uj72VV2w3LPay3cmw2c2vxM5iV

Copy Address

[USDT_TRX] TETHER TRC-20 ADDRESS FOR PAYMENT



TVjXLVLXKHnvR8YRNHKMStFmLuriSwynB1

Copy Address

Chat

We communicate only in English

You 8 days ago

hello

Reply

You 8 days ago

Target
hello

1

Reply

panda 8 days ago

Initial Leaking of Data

Apache Server status exposed

Leaking **URLs** / server **IP** address

Leaking **tokens** to check other people messages

You 8 days ago

hello

⇌ Reply

```
POST /chats/sendmsg HTTP/1.1
Host: wtyafjyhwqrgo4a45wdvvwhen3cx4euie73qvlhkhvrexljjoyuklaad.onion
Content-Length: 54
Accept: */*
X-Requested-With: XMLHttpRequest
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://wtyafjyhwqrgo4a45wdvvwhen3cx4euie73qvlhkhvrexljjoyuklaad.onion
Referer: http://wtyafjyhwqrgo4a45wdvvwhen3cx4euie73qvlhkhvrexljjoyuklaad.onion/d
ashboard/DEMDRcXdfNoTpKMKKwsE
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9
Cookie: PHPSESSID=b86f1b26891a51f6871e0d70018c6359
Connection: keep-alive

reply_id=37129&targetID=OCB8EBFA70F48DB2BD477BD1&msg=1
```

Mallox

IDOR on reply_id parameter

Parameter was an **incremental** integer

For loop to the rescue!

Mallox

Boss

I work according to my own schedule, which does not change at anyone's request. As soon as I see the payment I will send a decryptor

Boss

I have a well maintained business that is in no way going to be disturbed by any moron that could not secure their machines. The price is final and non negotiable. You should pay or data will be released

Boss

I was on Special K for a couple of days so fuck off and do the work I am paying to do.

Mallox

Panda

Boss, the budget for this client is only \$1500. Do you agree? They don't need much data

Jessica

Boss is not on a good mood today. Let's be careful

Mallox

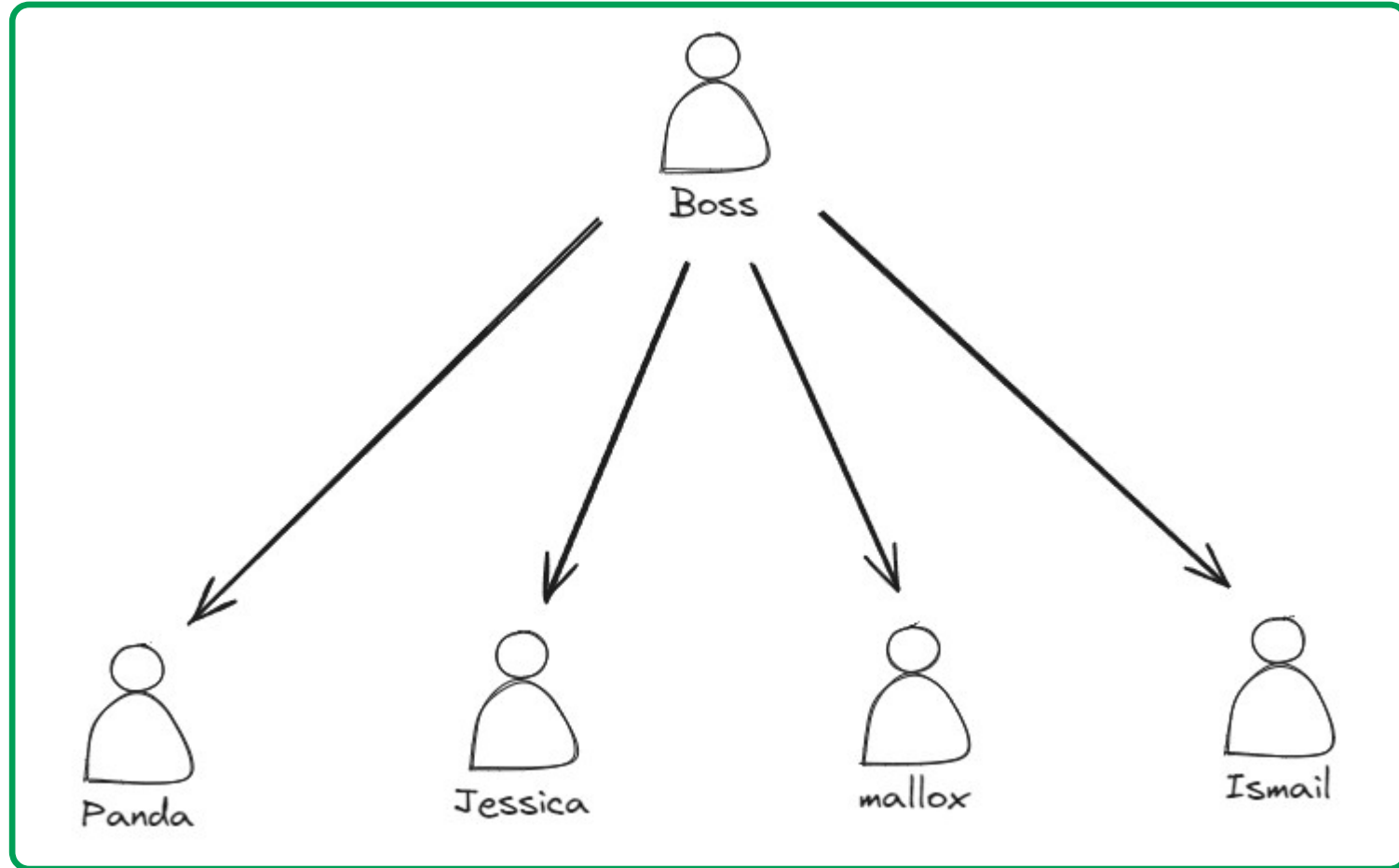
Boss

Here is the decryptor for [REDACTED]

Boss

Here is the decryptor for [REDACTED]

Mallox



Mallox

BOT 37 seconds ago ✓

The chat was disabled for OCB8EBFA70F48DB2BD477BD1

↔ Reply

```
HTTP/1.1 200 OK
Server: nginx/1.24.0
Date: Mon, 22 Jul 2024 19:19:06 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
X-Powered-By: PHP/7.4.33
Expires: Thu, 19 Nov 1981 08:52:00 GMT
Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache
Content-Length: 104

<script>
  notifications("Message was not delivered", 'error');
</script>
```

Mallox

Got **internal knowledge** of the team

Got some **decryptors**

Once the admin got up it was fixed

Mallox





BlackCat (ALPHV)

BlackCat AKA Noberus and **ALPHV**

First appearance in **2021**

RaaS

Rust-based **malware** and server-side **C2**

Triple extortion scheme



BlackCat (ALPHV)

Rustloader which was researched by **Bitdefender Labs**

Targeting **Mac** only

Developed in **Rust**

C2 used on all of **BlackCat** Malware

BlackCat (ALPHV)

ANTI-MALWARE RESEARCH • ⌚ 9 min read • 📌

New macOS Backdoor Written in Rust Shows Possible Link with Windows Ransomware Group



Andrei LAPUSNEANU
February 08, 2024

Promo Protect all your devices, without slowing them down.
[Free 30-day trial](#)

C&C URLs

maconlineoffice.com

193.29.13.167

88.214.26.22

<https://serviceicloud.com>

BlackCat (ALPHV)

Checked **all** C2 URLs

One of them was **online** but was just returning **404**

Added in the loop of **continuous** scanning

2 days later it downloaded the **documentation**

FastAPI - ReDoc

servicecloud.com/redoc#tag/Client

Search...

Gateway >

Tasks >

Client ▾

GET Get Bots

PUT Set Note

Client

Get Bots

Responses

> 200 Successful Response

Set Note

PATH PARAMETERS

bot_id required integer (Bot Id)

REQUEST BODY SCHEMA: application/json

note required string (Note)

Responses

> 200 Successful Response

> 422 Validation Error

GET /client/bots ▾

Response samples

200

Content type application/json

null Copy

PUT /client/set-note/{bot_id} ▾

Request samples

Payload

Content type application/json

{ "note": "string" } Copy

Response samples

200 422

Content type application/json



serviceicloud.com/redoc#tag/Tasks

Search...

Gateway

Tasks

POST Create Task

POST Upload File

GET Get Task Result

DEL Delete By Timeout

Client

Tasks

Create Task

REQUEST BODY SCHEMA: application/json

name
required

string (Name)

params

string (Params)

bot_id
required

integer (Bot Id)

Responses

> 200 Successful Response

> 422 Validation Error

Upload File

REQUEST BODY SCHEMA: multipart/form-data

file
required

string <binary> (File)

POST /tasks/create

Request samples

Payload

Content type
application/json

```
{
  "name": "string",
  "params": "string",
  "bot_id": 0
}
```

Copy

Response samples

200 422

Content type
application/json

null

Copy

POST /tasks/upload_file

Response samples

422

BlackCat (ALPHV)

Endpoints were **not available**

Automate check and put it in the loop

Wait...

BlackCat (ALPHV)



BlackCat (ALPHV)

Extracted **197 commands** in **2 minutes** on a **4-hour window**

```
HTTP/1.1 200 OK
Server: nginx/1.15.12
Date: Sun, 11 Feb 2024 21:42:21 GMT
Content-Type: application/json
Content-Length: 135
Connection: keep-alive

{
  "params": "files/635281ed-4048-47a4-94f7-c6d769c79371.file",
  "name": "upload",
  "completed": true,
  "bot_id": 1,
  "id": 4,
  "result": "U3VjY2Vzcw=="
}
```


BlackCat (ALPHV)

```
HTTP/1.1 200 OK
Server: nginx/1.15.12
Date: Sun, 11 Feb 2024 21:42:21 GMT
Content-Type: application/json
Content-Length: 135
Connection: keep-alive
{
  "params": "zip -r env.zip xxxx-environments",
  "name": "shell",
  "completed": true,
  "bot_id": 5,
  "id": 29,
  "result": "YWRkaW5nOiBiaXRzby11bnZpcm9ubWVudHMv"
```

BlackCat (ALPHV)

Identified 4 companies

All of them crypto(currency) related

2 of the were unicorns

Notified and acknowledged the issue

None of them was ransomed

BlackCat (ALPHV)

Got **all info** and understood the lateral movement

Stopped the whole campaign

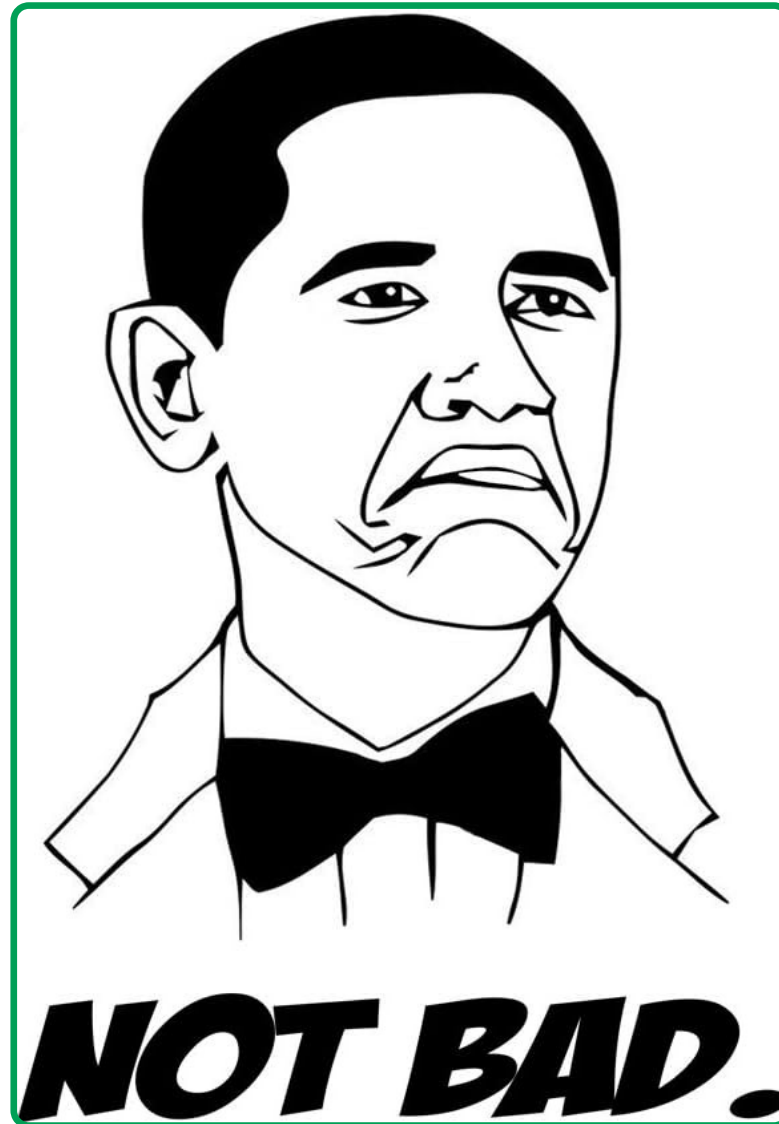
4 companies **not** ransomed

Financial hit

BlackCat (ALPHV)



BlackCat (ALPHV)



Everest

Active since late **2020**

High profile targets

Pivoted to **Initial Access Broker**

Re-pivoted to just **data leak**

Graded as “highly sophisticated” by multiple researchers

Everest

Everest Ransomware Group

[Home](#) [About](#)



SH Pension

Company has the last 24 hours to contact us using the instructions left. In case of silence, all data will be published. Total amount of stolen data : 100 GB <https://www.shpension.se/>

//

STUDIO NOTARILE BUCCI – OLMI

Company must contact us using the instructions in next 3 days. Total amount of stolen data: 400 GB <https://studionotarilebucciolmi.it>

Everest

/wp-includes/css/dist/block-library/style.min.css?ver=5.5.3

Not secure ransomcmou6mnbquqz44ewosbjk3o5qjsl3orawojexfook2j7esad.onion/wp-login.php



Username or Email Address

Password

☐ Remember Me

Not secure ransomcmou6mnbquqz44ewosbjk3o5qjsl3orawojexfook2j7esad.onion/readme.html



Semantic Personal Publishing Platform

First Things First

Welcome. WordPress is a very special project to me. Every developer and contributor adds something unique to the mix, and together we create something beautiful that I'm proud to be a part of. Thousands of hours have gone into WordPress, and we're dedicated to making it better every day. Thank you for making it part of your world.

— Matt Mullenweg

Installation: Famous 5-minute install

1. Unzip the package in an empty directory and upload everything.
2. Open [wp-admin/install.php](#) in your browser. It will take you through the process to set up a wp-config.php file with your database connection details.
 1. If for some reason this doesn't work, don't worry. It doesn't work on all web hosts. Open up wp-config-sample.php with a text editor like WordPad or similar and fill in your database connection details.
 2. Save the file as wp-config.php and upload it.
 3. Open [wp-admin/install.php](#) in your browser.

Everest

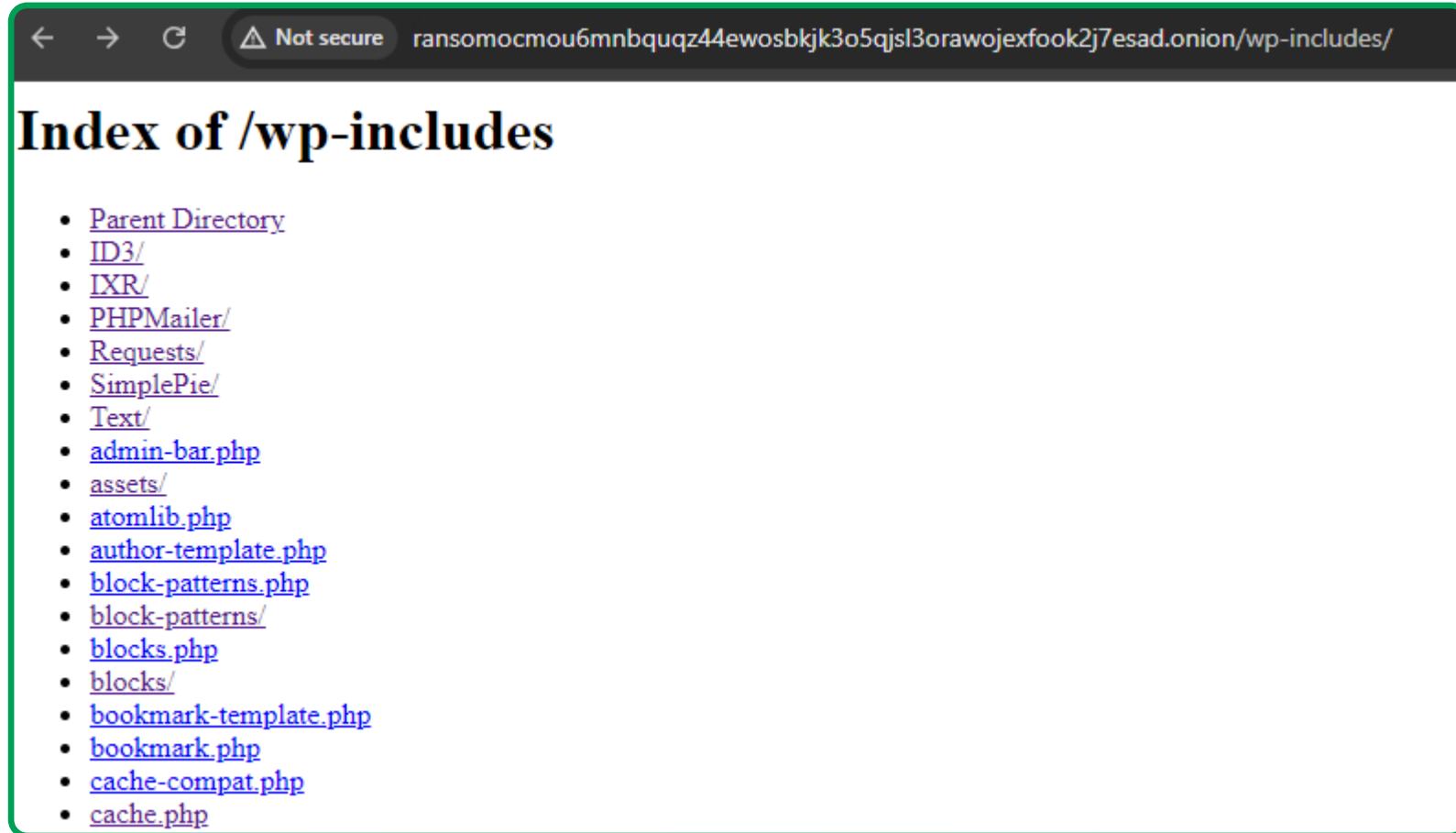
```
[+] WordPress version 5.5.3 identified (Insecure, released on 2020-10-30).  
| Found By: Rss Generator (Passive Detection)  
| - http://ransomocmou6mnbquqz44ewosbkjk3o5qjsl3orawojexfook2j7esad.onion/feed/,  
| - http://ransomocmou6mnbquqz44ewosbkjk3o5qjsl3orawojexfook2j7esad.onion/commen  
tor>  
|  
| [!] 42 vulnerabilities identified:  
|  
| [!] Title: WordPress 4.7-5.7 - Authenticated Password Protected Pages Exposure  
| Fixed in: 5.5.4
```

Everest

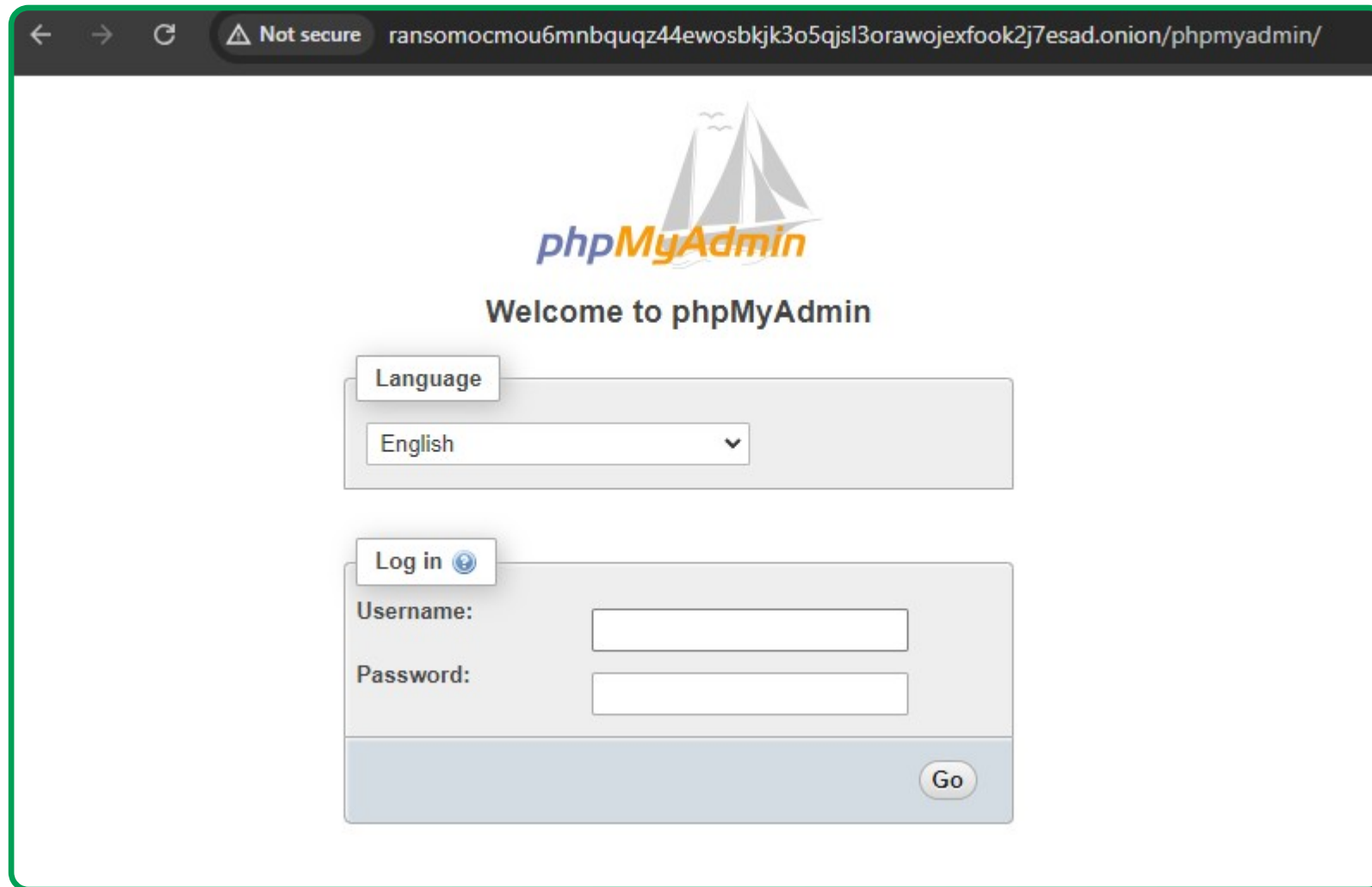
(!) Fatal error: Call to undefined function _deprecated_file() in C:\Program Files\VertrigoServ\www\wp-includes\rss-functions.php on line 9				
Call Stack				
#	Time	Memory	Function	Location
1	0.0006	137240	{main}()	...\rss-functions.php:0

- php_error reporting on
- **Windows** machine
- **VertrigoServ** hosting

Everest



Everest



A screenshot of a web browser displaying the phpMyAdmin login page. The browser's address bar shows a URL starting with 'ransomocmou6mnbquqz44ewosbkjk3o5qjsl3orawojexfook2j7esad.onion/phpmyadmin/'. The page features the phpMyAdmin logo, which includes a stylized sailboat and the text 'phpMyAdmin'. Below the logo, it says 'Welcome to phpMyAdmin'. There are two main sections: a language selection dropdown currently set to 'English', and a login form with fields for 'Username:' and 'Password:', followed by a 'Go' button. A 'Log in' link with a question mark icon is also present above the login form.

← → ↻ ⚠ Not secure ransomocmou6mnbquqz44ewosbkjk3o5qjsl3orawojexfook2j7esad.onion/phpmyadmin/


Welcome to phpMyAdmin

Language

English ▼

Log in ?

Username:

Password:

Go

Everest

MySQL's default user name is "root" and password is "vertrigo". In addition, SQLiteManager can use both "username" and "password" as "admin" "data" "guest".

Everest



Everest



Everest



Everest

WordPress

Everest Ransomware Group

10

0

New

CleanTalk 30

UpdraftPlus

Howdy, admin

Screen Options

Help

Dashboard

Home

Updates 10

Posts

Media

Pages

Comments

Appearance

Plugins 6

Users

Tools

Settings

Duplicator

Collapse menu

WordPress 6.6.1 is available! [Please update now.](#)

Please enter the Access Key in [Anti-Spam by CleanTalk](#) plugin to enable spam protection!

Thank you for installing UpdraftPlus!

Dismiss (for 12 months)

Super-charge and secure your WordPress site with our other top plugins:
[UpdraftPlus Premium](#): For personal support, the ability to copy sites, more storage destinations, encrypted backups for security, multiple backup destinations, better reporting, no adverts and plenty more, take a look at the premium version of UpdraftPlus - the world's most popular backup plugin. [Compare with the free version](#) / [Go to the shop.](#)
[UpdraftCentral](#) is a highly efficient way to manage, update and backup multiple websites from one place.
[WP-Optimize](#): Makes your site fast and efficient. It cleans the database, compresses images and caches pages for ultimate speed.
[Subscribe](#): The WordPress subscription extension for WooCommerce store owners.
[Free Newsletter](#): UpdraftPlus news, high-quality training materials for WordPress developers and site-owners, and general WordPress news. You can de-subscribe at any time.
More quality plugins : [Premium WooCommerce plugins](#) | [Free two-factor security plugin](#)

Dismiss (for 12 months)



Dashboard

Welcome to WordPress!
We've assembled some links to get you started:

Dismiss

Get Started

Customize Your Site

or, [change your theme completely](#)

Next Steps

[Write your first blog post](#)

[Add an About page](#)

[Set up your homepage](#)

[View your site](#)

More Actions

[Manage widgets](#)

[Manage menus](#)

[Turn comments on or off](#)

[Learn more about getting started](#)

PHP Update Required

WordPress has detected that your site is running on an insecure version of PHP.

Quick Draft

Title

Everest

```
Administrator@WIN-50DCFIGQRP3:C:\Program Files\VertrigoServ\www\wp-content\plugins# ipconfig
```

```
Ⓜ áääâ©©² ²à©â©²©« IP Ⓜ«ī Windows
```

```
Ethernet adapter Ethernet:
```

```
DNS-áääää²á ²©Ⓜ²«īç¥²ī . . . . . :  
Ⓜ©² «îë© IPv6- Ⓜà¥á² « . . . . : fe80::b9c0:4fb7:8f3d:1052%12  
IPv4- Ⓜà¥á. . . . . : 92.38.241.244  
Ⓜ á² ²©Ⓜá¥â² . . . . . : 255.255.255.0  
Ⓜá©ç©© è«î§. . . . . : 92.38.241.1
```

```
Ⓜã¥«îë© Ⓜ ²â¥à isatap.{66FC55AB-B4E8-43B6-97D0-5E7F248C9BF3}:
```

```
Ⓜ©áâ©ī²¥ áà¥Ⓜë. . . . . : Ⓜà¥Ⓜ ²¥à¥Ⓜ ç² ¥Ⓜ©áää² .  
DNS-áääää²á ²©Ⓜ²«īç¥²ī . . . . . :
```

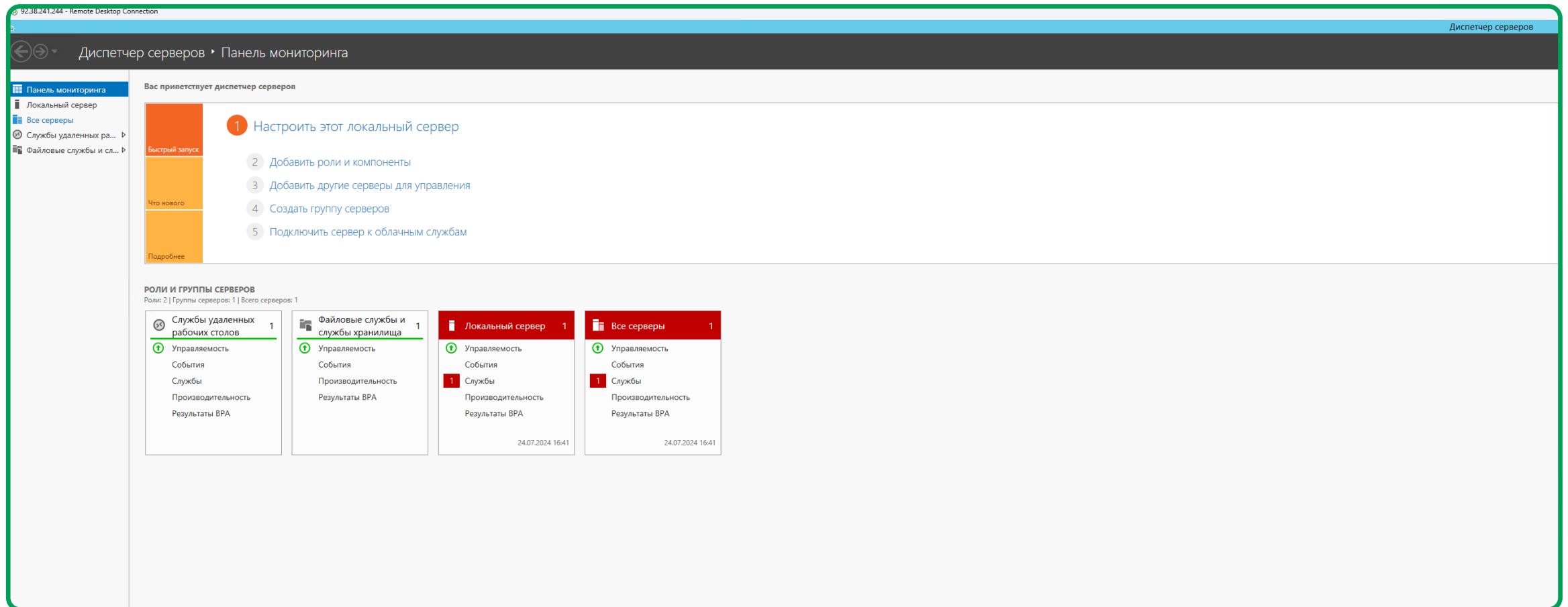
```
Ⓜã¥«îë© Ⓜ ²â¥à 6T04 Adapter:
```

```
DNS-áääää²á ²©Ⓜ²«īç¥²ī . . . . . :
```

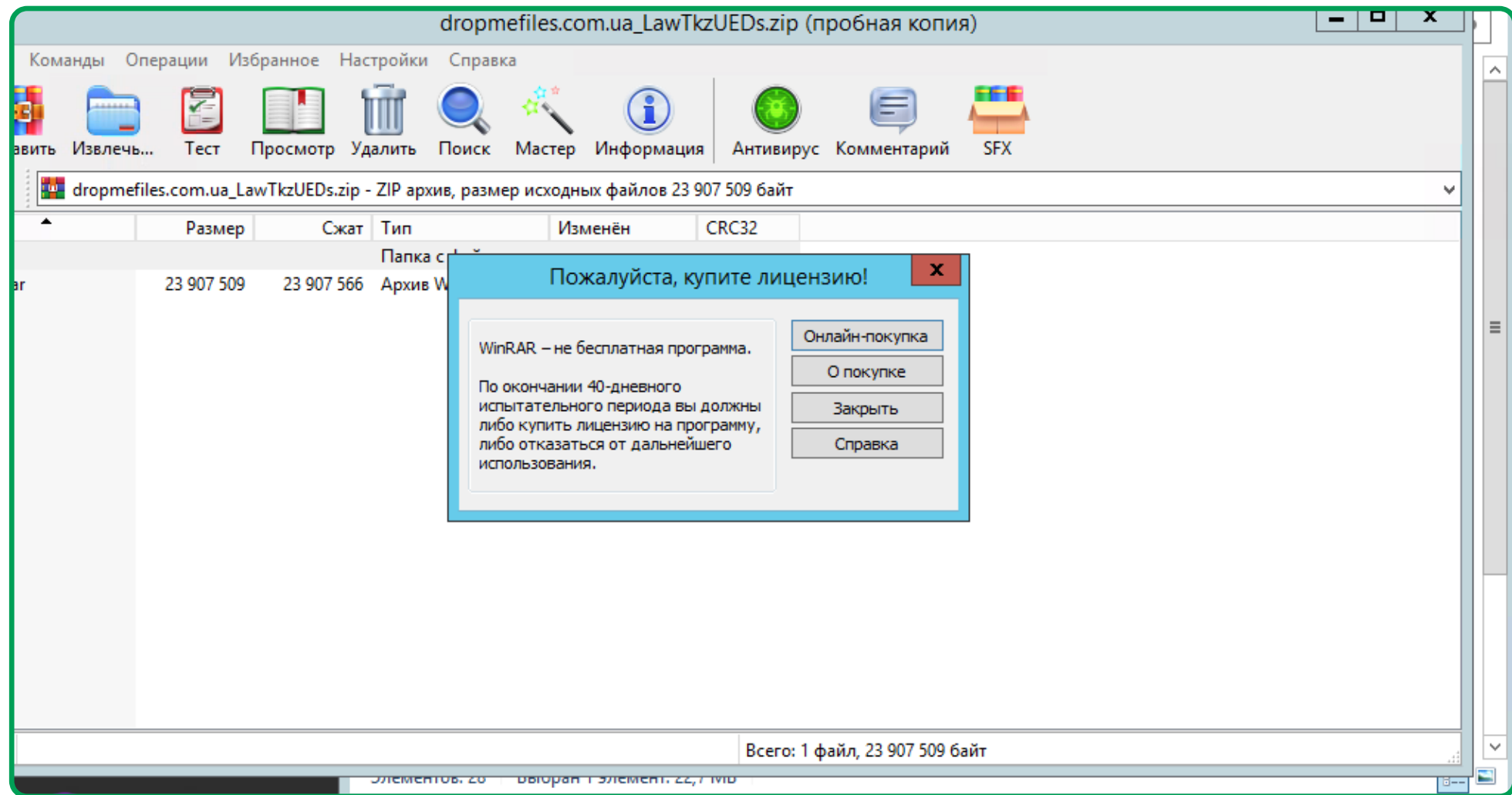
Everest

```
Administrator@WIN-50DCFIGQRP3:C:\Program Files\VertrigoServ\www# whoami  
win-50dcfigqrp3\administrator
```

Everest



Everest



Everest

Database export

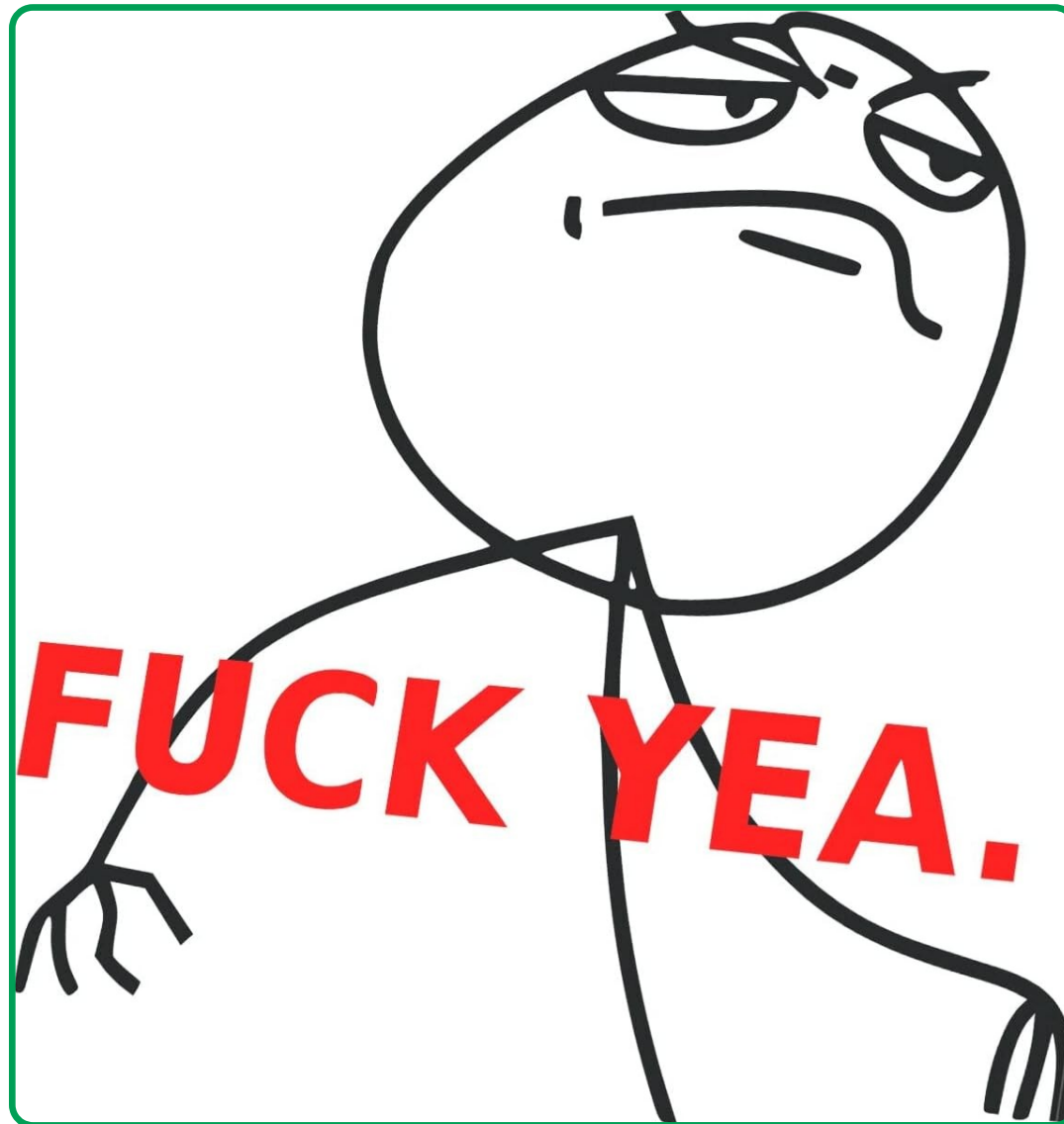
Username extract

RCE on the server

Got hold of their onion secret keys

Lots of logs to analyze

Everest



Black Angels

Emerged in May **2022**

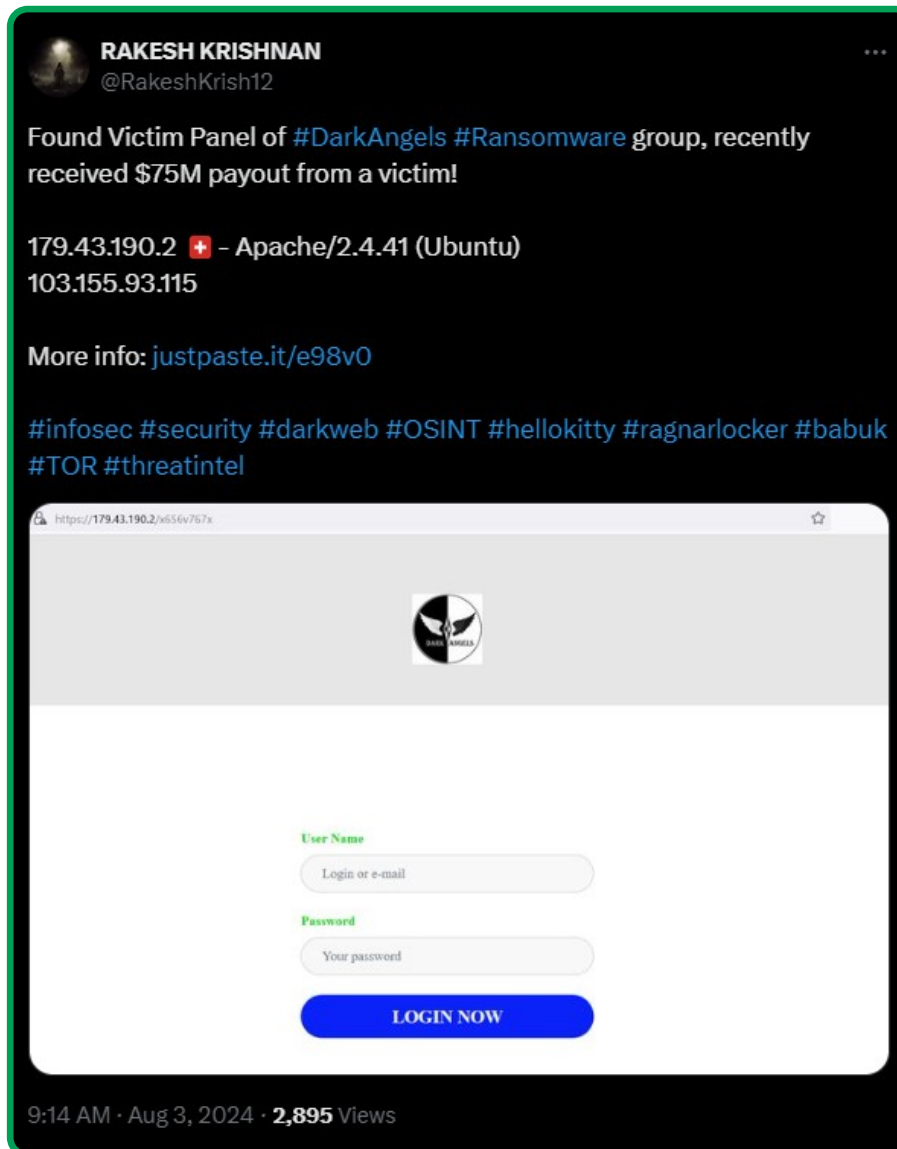
High profile targets

Babuk based

Linux / ESXI focused

Received a record **\$75M** ransom

Black Angels



Black Angels

```
← → ↻ ⚠ Not secure 179.43.190.2/.env

# In all environments, the following files are loaded if they exist,
# the latter taking precedence over the former:
#
# * .env                contains default values for the environment variables needed by the app
# * .env.local          uncommitted file with local overrides
# * .env.$APP_ENV       committed environment-specific defaults
# * .env.$APP_ENV.local uncommitted environment-specific overrides
#
# Real environment variables win over .env files.
#
# DO NOT DEFINE PRODUCTION SECRETS IN THIS FILE NOR IN ANY OTHER COMMITTED FILES.
#
# Run "composer dump-env prod" to compile .env files for production use (requires symfony/flex >=1.2).
# https://symfony.com/doc/current/best_practices.html#use-environment-variables-for-infrastructure-configuration

###> symfony/framework-bundle ###
APP_ENV=prod
APP_SECRET=d4b817d4c6966202b1bdfb57a46e4d7c
TRUSTED_PROXIES=127.0.0.0/8,10.0.0.0/8,172.16.0.0/12,192.168.0.0/16
TRUSTED_HOSTS='^(localhost|example\.com)$'
###< symfony/framework-bundle ###

###> symfony/mailer ###
MAILER_DSN=smtp://localhost
###< symfony/mailer ###

###> doctrine/doctrine-bundle ###
# Format described at https://www.doctrine-project.org/projects/doctrine-dbal/en/latest/reference/configuration.html#connecting-using-a-url
# For an SQLite database, use: "sqlite:///kernel.project_dir%/var/data.db"
# For a PostgreSQL database, use: "postgresql://db_user:db_password@127.0.0.1:5432/db_name?serverVersion=11&charset=utf8"
# IMPORTANT: You MUST configure your server version, either here or in config/packages/doctrine.yaml
DATABASE_URL=mysql://chat2:JjLKHkJHkJHkGG1134@localhost:3306/chat?serverVersion=5.7
###< doctrine/doctrine-bundle ###

###> symfony/messenger ###
# Choose one of the transports below
# MESSENGER_TRANSPORT_DSN=amqp://guest:guest@localhost:5672/%2f/messages
# MESSENGER_TRANSPORT_DSN=doctrine://default
# MESSENGER_TRANSPORT_DSN=redis://localhost:6379/messages
###< symfony/messenger ###

REGISTER=0;
```

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`.env` file exposure

Internal URLs exposed

Some error logs exposed

1 message mentioning 5.5M payment exposed

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ETHICAL DILEMMAS



Conclusions



The logo for the UXUG Five Horsemen Tour features a central figure, a knight in armor, riding a horse and holding a sword. This central figure is flanked by four other figures, each riding a horse and holding a banner. The banners are labeled 'SPYWARE', 'RANSOMWARE', 'BOTNET', and 'STEALER'. The central figure's horse is labeled 'BOTNET'. The entire logo is set against a black background with a green border.

UXUG

FIVE HORSEMEN TOUR

Spyware - Bsides London 2022

Stealer / Botnet - Defcon 31

Ransomware - Defcon 32 / Blackhat 2024



THANK YOU